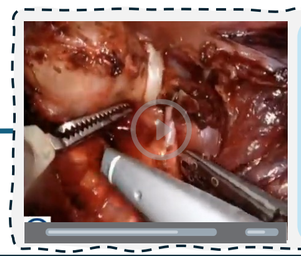
Problem Statement

Pattern matching responses based on standard labeled datasets!

The phase is *gallbladder dissection*, and the tool is a *grasper*

prior surgical VLMs



Q: Why did the surgeon decide to sacrifice the adjacent branch?

SureonVLM-R1 (ours)

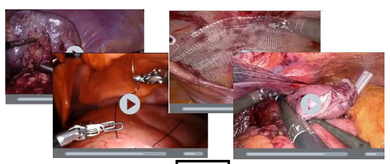
<think>The team encountered a large lymph node adjacent to the arterial branch, and the node was too bulky to preserve without compromising the vessel. The branch was sacrificed to complete nodal clearance safely </think> <answer> **The adjacent arterial branch was too bulky to preserve without compromising the vessel** </answer>

Semantically the same

GT: An enlarged interlobar lymph node requiring complete en bloc removal

SUREON Data Synthesis

Narrated Surgical Videos



Transcripts

Specialized generator agent

① Identify Semantic Grounding Moments (SGMs)

② Generate structured samples

QUESTION (Q)

Options (O_1) … Options (O_n)

Answer (A)

CoT * (R)

*for some QA types

Grounded in transcript

Referring to present scene

Task-specific criteria

Specialized filtering agent

x12 Specialized agents

12 Question types

206.8K Video clip conversations

354 Test expert-verified conversations

Training

Stage 1: Supervised Fine-tuning (SFT)

SUREON dataset → Q O

Standard datasets → Q O

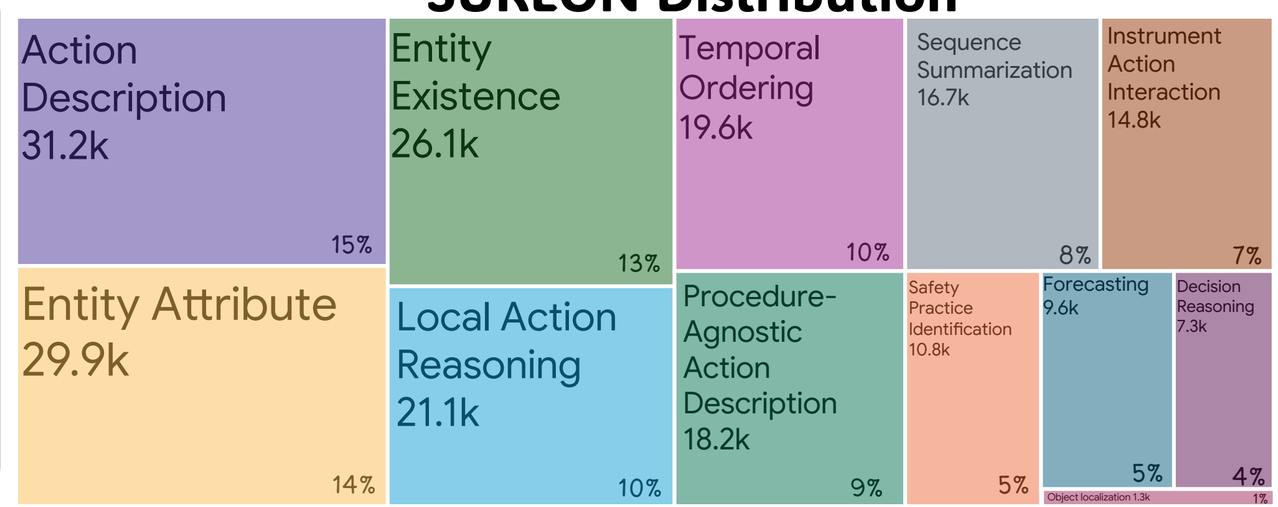
SureonVLM → Answer, R^*

clips frames

*for some QA types

Stage 2: Reinforcement Learning

SUREON dataset → Q O → SureonVLM

group completions → Rewards → GRPO

SUREON Distribution